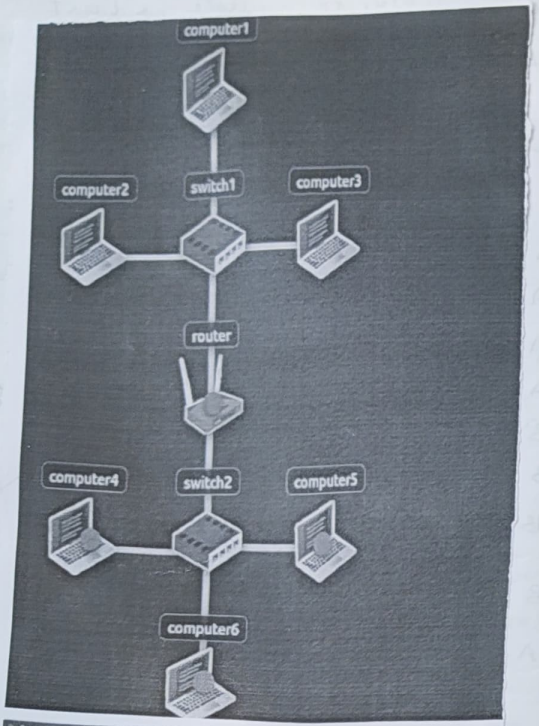
10/9/24 Ex No 8 : NMAP to Discover Live Hosts using NMap Scans (ARP, ICMP, TCP/UDP) on TryHackMe Platform

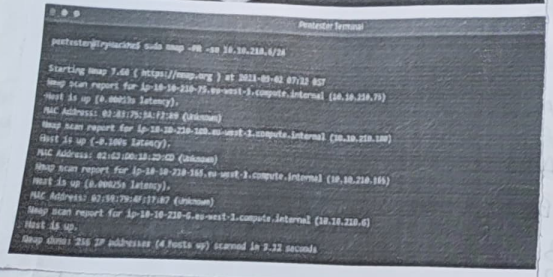
(a)

AIM:

To discover Live Hosts using NMap Scans (ARP, ICMP, TCP/UDP) on the TryHackMe Platform

Room Link : https://tryhackme.com/room/nmapol





① INTRODUCTION to Nmap

Nmap is a free and open-source network scanning tool used to discover active systems and identify running services on a network.

Before performing port scanning, it is useful to determine which systems are actually online. Scanning offline systems waste time and can create unnecessary network traffic.

Nmap provides different methods for live host discovery, including

* ARP scan
* ICMP scan
* TCP SYN ping
* TCP Ack ping
* UDP ping

② SUBNETWORKS:

A network segment is a group of computers connected through a common medium such as an Ethernet switch/ Wifi Access point

Ex:

Subnet	Subnet Mask	Approx Hosts
/16	255.255.0.0	≈65000
/24	255.255.255.0	≈250

ARP based discovery works only within the same subnet because ARP operates at the link layer and its requests cannot cross routers

③ Enumerating Target

Nmap needs to know IP address. It can be specified as

- Single IP address 10.10.10.5
- Range of IP address - 10.10.10.5-10
- subnet : 10.10.10.0/24
- Disable DNS resolution. nmap -n TARGETS

This target enumeration helps define the scope of the scan

④ Live Host Discovery

The experiment covers 4 protocols

ARP - Link Layer
ICMP - Network Layer
TCP - Transport Layer
UDP - Transport Layer

⑤ ARP Host Discovery

* ARP requests are used to find live hosts on the same subnet

* A live host respond with an ARP reply.

* ARP cannot cross routers.

* nmap -PR -sn TARGETS performs an ARP based host discovery without port scanning.

6. ICMP Host Discovery

* ICMP Echo uses type 8 and type 0.

* Nmap options : -PE

* ICMP Timestamp ecen : -PP

* ICMP may not always work because firewalls can block ICMP requests

7. TCP SYN Ping:

- sends a TCP packet with SYN flag set
- A open port may respond with SYN/ACK.
- A closed port may respond with RST
- Nmap option :- PS

8. TCP ACK ping:

- sends a TCP packet with ACK flag set
- Normally uses port 80 by default
- Nmap option : -PA
- A target may respond with a RST, indicating the host is active
- TCP ACK Ping requires a privileged account.

9. UDP Ping:

- UDP packets can be used to determine whether a host in active.
- A packet sent to a closed UDP port may produce an ICMP "Port Unreachable" response.
- This response indicates that the target host and reachable.

10. Reverse DNS lookup:

- Nmap normally performs reverse-DNS lookups for online hosts
- -n → disables DNS resolution
- -R → performs reverse-DNS lookup
- A specific DNS server can be also specified using ---dns-server

DNS server

	Purpose	option
1.	Ping host discover only	-sn
2.	ARP scan	-PR
3.	ICMP Echo	-PE
4.	ICMP Timestamp	-PP
5.	ICMP Address mask	-PM
6.	TCP SYN Ping	-PS
7.	TCP ACK Ping	-PA
8.	Disable DNS lookup	-n
9.	Force reverse DNS	-R
10.	List targets without scanning	-sL
11.	Read targets from file	-iL

Result: The Nmap live Host Discovery experiment was successfully studied on TryHackMe platform. Live hosts were identified using ARP, ICMP, TCP, ACK and UDP techniques and the corresponding Nmap options were understood.